Using Claude Code for Data Analysis

Directing, inspecting, and learning from agent-assisted work in R

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The argument of this workshop

Coding agents are often effective at translating well-specified statistical decisions into runnable R code.

They are much less reliable when they must decide what should be estimated, which assumptions are defensible, or what the result means.

Effective use therefore depends on statistical expertise, explicit direction, and verification.

What you will learn

By the end of the workshop, you should be able to:

1. Explain what a coding agent does when it analyzes data with R.
2. Write an analysis brief that fixes the goal, target, task, and outputs.
3. Trace a claim back through its sample, estimator, code, and evidence.
4. Recognize a consequential choice that the agent supplied.
5. Decide when to proceed, revise, stop, or seek statistical expertise.

The objective is not to avoid learning R or statistics. It is to use an agent without surrendering the decisions that make an analysis meaningful.

What research evaluates

"Can an LLM do statistics?" combines several different tasks.

Task	What success means
Select a named test	Map a well-described design to a known procedure
Write statistical code	Produce executable syntax that implements a specification
Reproduce an analysis	Recover a known result from supplied materials
Design an analysis	Choose a target, sample, model, and diagnostics
Interpret evidence	Connect estimates and assumptions to a bounded claim
Complete a workflow	Coordinate files, tools, debugging, validation, and reporting

Bounded tasks can go very well

Sebo and Wang tested GPT-5, Claude, and DeepSeek on 32 statistical examples with standardized prompts.

Evaluation	Result
Identify the statistical test	32/32 for every model in both rounds
Generate the Stata command	87.5–96.9% no/low-risk deviations
Repeat the command across rounds	93.8–100% reproducibility

The benchmark supplied canonical problems, a reference answer, a fixed prompt, and a narrow output.

Sebo & Wang (2026). The study intentionally excluded assumption checking and result interpretation.

Statistical code is useful—and uneven

Song and colleagues asked GPT-3.5, GPT-4, and Llama 3.1 to write SAS code for 207 human-verified statistical tasks.

- Models often produced syntactically correct and readable code.
- Performance weakened on tasks requiring deeper domain understanding.
- Executable-looking programs could still contain redundant analyses or incorrect results.
- Non-expert users were least equipped to recognize those errors.

Code generation is a real capability. Statistical validity is a separate outcome.

Song et al. (2025). Evaluation by statistics faculty and PhD researchers.

Frontier coding agents can reproduce known work

Alizadeh and colleagues tested Claude Opus 4.6 in Claude Code and GPT-5.3-Codex on 221 social-science reproduction tasks.

Agent	Task-level accuracy	Entire-paper accuracy
Claude Code	93.4%	78.0%
Codex	62.1%	35.8%

These tasks began with complete, executable replication materials and a known result. Claude also repaired paths and dependencies successfully.

Alizadeh et al. (2026), SocSci-Repro-Bench; three runs per paper.

Open-ended statistical analysis remains difficult

StatABench evaluates both closed questions and open-ended modeling with tools.

Benchmark	Best reported performance
404 closed statistical questions	GPT-5.1: 68.6% correct
30 open-ended modeling problems	Top framework: 61.86 of 100 points

The persistent failures involved tool-grounded reasoning, methodological decisions, and end-to-end modeling—not merely R or Python syntax.

Zhu et al. (2026). Closed questions span 18 topics; open tasks come from modeling competitions.

Full workflows add another failure layer

DSAgentBench asks agents to complete 275 multi-stage data-science tasks in real computer environments.

56.7%

task success for the strongest tested agent, Claude 4.6 Sonnet

The difficult parts included grounding in intermediate results, coordinating tools, navigating the operating system, and maintaining a correct multi-step plan.

Rahman et al. (2026). Correctness was execution-tested, not judged from code appearance alone.

A plausible report can conceal analytic selection

Miao and colleagues held the data and question fixed while varying an agent's stated prior belief.

- Persona prompts produced divergent—and sometimes opposing—conclusions.
- Agents reproduced 72% of the ideological gap seen among 42 human teams.
- 86% of agent reports passed independent AI review.
- 78% passed review by a majority of human experts.

The problem was often selective exploration among defensible paths, not an obviously broken final model.

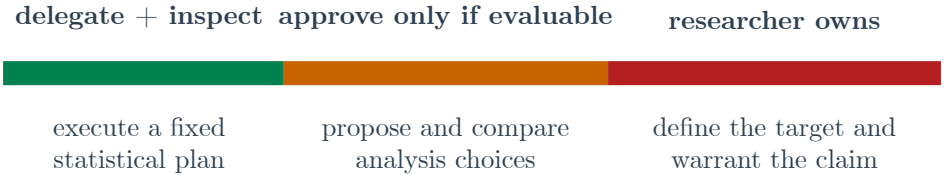
Miao, Pritchard, & Zou (2026), using Claude Code with Sonnet 4.6.

What we know about doing statistics with agents

Strongest evidence	Remaining uncertainty
A clear specification can be translated into working code.	Did we specify the right quantity and population?
Saved scripts can make a result computationally reproducible.	Does the script implement a defensible analysis?
Agents can repair many ordinary execution failures.	Did a repair change the sample, method, or meaning?
Agents can propose multiple plausible models quickly.	Why was this path selected and others omitted?
Agents can write fluent interpretations.	Does every clause stay within the design and assumptions?

Good instruction improves execution. It does not remove uncertainty about the statistical decisions.

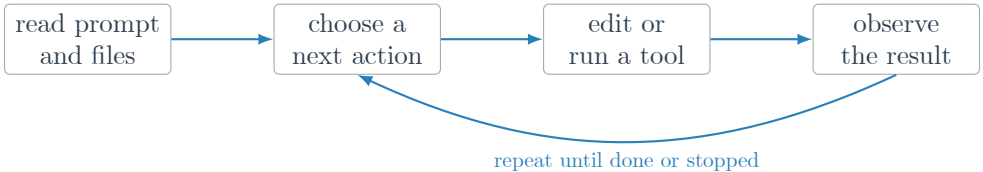
The practical capability boundary



Delegation should stop at the edge of your ability to evaluate the consequence of a choice.

Claude Code is not just a chat window

A coding agent can inspect a project, edit files, run R, observe errors, and revise its work.



That loop is useful because code is executable and inspectable. It is risky because the agent can compound an early statistical assumption across many files.

Give the analysis a project structure

```
claude-data-analysis/
   data/           # original inputs; read-only
   exercises/      # briefs and participant instructions
   scripts/        # approved R code and run outputs
   workshop/       # slides and assets
```

Ask the agent to use relative paths and to write new artifacts only under the designated output folder.

```
d <- read.csv("data/salaries/salaries.csv")
write.csv(results, "scripts/we-do/model-results.csv", row.names = FALSE)
```

Project structure is a technical control: it makes inputs, actions, and outputs easier to inspect.

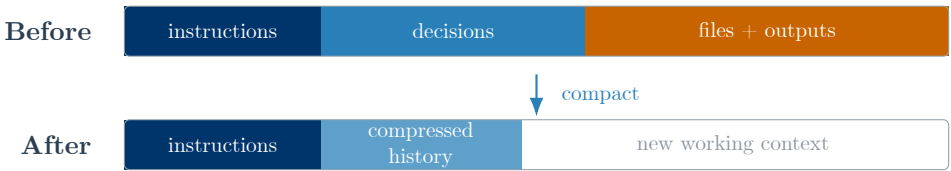
A context window fills during the analysis



More context is not the same as better evidence.

The model works from a finite record of instructions, conversation, file content, and recent tool output.

Compaction preserves a summary, not the full history



Put consequential decisions in durable files—an approved brief, plan, script, result table, and decision log—then tell the agent to reopen them after compaction.

Skills and subagents are workflow aids

Mechanism	Useful for	Does not establish
Project instructions	Persistent local rules and file conventions	That the rules are statistically sound
Skill	A reusable procedure, checklist, or output format	That the procedure fits this scientific question
Subagent	A bounded second pass, code review, or alternative proposal	Independent evidence or independent judgment

Use these mechanisms when repetition or context management is the problem.

Do not use them to manufacture consensus about a choice you cannot evaluate.

A prompt should function as an analysis brief

The useful unit is not a clever sentence. It is a prospective record of the intended analysis.

Part	What it fixes before results are seen
Goal	Motivation, question, hypothesis, target quantity
Data	Unit, population, provenance, coding, time, access boundary
Task	Estimator, sample, assumptions, diagnostics, alternatives
Output	Files, tables, figures, precision, run record
Decision rights	What the agent may execute, propose, or change only after approval
Stop conditions	Evidence that pauses, reopens, or ends the analysis

Good prompting is statistical specification plus project management.

Goal: name the quantity before the method

Compare these requests:

(1) *Does college increase political participation?*

(2) *Estimate the average difference in the 0–8 participation index that would be expected if everyone in this panel attended college versus if no one attended, under stated identification assumptions.*

Goal: name the quantity before the method

Compare these requests:

> ***(1)*** *Does college increase political participation?*
>
> ***(2)*** *Estimate the average difference in the 0–8 participation index that would be expected if everyone in this panel attended college versus if no one attended, under stated identification assumptions.*

Request (2) names a target quantity. It still requires decisions about population, timing, identification, and estimation.

Goal: name the quantity before the method

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> ***(1)*** *Does college increase political participation?*
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> ***(2)*** *Estimate the average difference in the 0–8 participation index that would be expected if everyone in this panel attended college versus if no one attended, under stated identification assumptions.*

Request (2) names a target quantity. It still requires decisions about population, timing, identification, and estimation.

Include the motivation and directional hypothesis—but state that the expected result must not guide specification search.

Data: tell the agent what one row means

A usable data section identifies:

- exact files and provenance;
- unit of observation and target population;
- treatment, outcome, covariates, identifiers, and time ordering;
- missing-value encodings and categorical-variable coding;
- read-only inputs and permitted output directory;
- known uncertainties the audit must resolve.

File names and variable labels are claims, not evidence. Ask the agent to compare documentation with the values in the data.

Task: specify the estimator and its checks

An estimator is a rule that maps data to an estimate—not merely a model name.

```
Fit yppnscal ~ college + approved 1965 covariates.
Predict every respondent's outcome with college = 1 and college = 0;
report the average of the differences with an HC2 95% interval.
Use all rows unless the audit finds a data defect.
Before fitting, report the sample size and treatment support.
Recompute the average 1-versus-0 contrast directly as a check.
```

If the method is unfamiliar, ask the agent to propose options and stop. Approval still requires knowledge or expert help.

Output: make the work inspectable

Request named artifacts instead of one polished response.

File	Purpose
`audit.md`	Data structure, coding, missingness, contradictions
`plan.md`	Frozen target, sample, estimator, diagnostics, stop rules
`analysis.R`	Executable implementation of the approved plan
`results.csv`	Machine-readable estimates, uncertainty, and sample size
`report.md`	Bounded interpretation linked to the result table
`decision-log.md`	Changes, approvals, reasons, and unresolved questions

Ask the agent to run `Rscript --vanilla analysis.R` and save the console output. A clean run is evidence of execution, not proof of validity.

Decision rights create checkpoints



Stop when evidence contradicts the plan, when changing the analysis would change the target, or when the required choice exceeds the researcher's expertise.

A complete compact prompt

```
Goal: Estimate the average effect of attending college on the 0--8 participation
index among respondents represented by the panel. If the files cannot establish
that college preceded the outcome, stop and propose a descriptive target.
Our hypothesis is positive, but do not search for it.

Data: Read data/kam-palmer/political_socialisation_data.csv and its codebook.
One row is one respondent. Treat inputs as read-only; write only under scripts/i-do/.

Task: First audit time ordering, coding, missingness, and derived variables.
Do not fit. Write audit.md and stop. After approval, propose an estimator and
checks in plan.md.

Output: After plan approval, write analysis.R, results.csv, report.md, and a run log.

Decision rights: I retain the target, adjustment set, sample, uncertainty
procedure, and interpretation.

Stop: Pause for contradictions, changed targets, unsupported causal language,
or a method I cannot evaluate.
```

I Do: the whole analysis, with decisions visible

Question: Does attending college increase later political participation?

Data: 1,254 respondents in the 1965–1973 Political Socialization Panel Study.

Treatment: attended college by 1973.

Outcome: number of political acts reported in 1973, from 0 to 8.

Initial target: the average causal effect for respondents represented by the panel.

The agent receives the brief, data, and codebook. Its first job is to audit and stop—not estimate.

I Do: why the audit comes first

The audit asks questions that can change the analysis before a model is fit:

- Are treatment and outcome ordered in time?
- Which variables are genuinely pre-treatment?
- Are variables duplicated or deterministic recodes?
- Does the documentation agree with the values?
- Do the data support the requested population and target?

The treatment and outcome were both recorded in the 1973 follow-up. The files do not establish that college attendance preceded every reported act.

That finding weakens the causal interpretation before any coefficient exists.

I Do: a polished label was not a valid check

The data contain `p1965_HHCollegePlacebo`, described as a placebo variable.

Direct inspection showed:

```
same <- with(d, p1965_HHCollegePlacebo == (p1965_EducHH >= 5))
table(same, useNA = "ifany")
# TRUE for all 1,247 non-missing rows; 7 rows are missing
```

It is a deterministic recode of household education—one of the adjustment variables—not an independent placebo outcome.

I Do: the human redefines what can be claimed

The audit does not support the requested causal target from the supplied information.

The approved fallback is narrower:

> *Estimate the adjusted difference in the participation index between college attendees and non-attendees: the average predicted difference, in index points, over this analysis sample.*

The agent may implement the model. The researcher owns:

- the change from a causal effect to an adjusted association;
- the adjustment logic and target population;
- the uncertainty procedure;
- the language used in the report.

Reframing is not failure. It is a statistical result of the audit.

I Do: execution follows the approved plan

Excerpt from `scripts/i-do/analysis.R`:

```
for (v in multi) a[[v]] <- factor(a[[v]])
rhs <- paste(c("college", covs), collapse = " + ")
hc2_omega <- function(residuals, diaghat, df) {
  ifelse(diaghat > 1 - 1e-8, 0, residuals^2 / (1 - diaghat))
}

fit <- lm(as.formula(paste("yppnscal ~", rhs)), data = a)
V <- vcovHC(fit, omega = hc2_omega)
est <- mean(predict(fit, a1) - predict(fit, a0))
se <- sqrt(V["college", "college"])
ci <- est + c(-1, 1) * qt(0.975, df_res) * se
```

`vcovHC(fit, type = "HC2")` returns `NaN` here: one respondent is alone in a category.

I Do: the handoff

In this analysis sample of 1,254 respondents, college attendees reported 0.65 more political acts on the 0–8 index than non-attendees, after adjusting for the approved 1965 covariates (HC2 95% interval 0.39 to 0.92). College attendance and participation were both recorded in 1973, so these data do not establish that college preceded the reported acts. The estimate is an adjusted association, not a causal effect.

I Do: 21 autonomous runs missed what the audit caught

Same data and question, run end to end with no checkpoints: seven versions of the brief, three runs each.

1 of 21

saw that the "placebo" is a recode of a covariate. 19 others ran it as a test.

0 of 21

asked whether college came before the reported acts. Both were recorded in 1973.

0 of 21

stepped back from a causal target: ATE (16), ATT (3), ATO (1), none stated (1).

The runs were diligent and consistent: 21 checked balance, every saved script reproduced its coefficient to 10^{-6}, and all estimates fell between 0.72 and 0.89.

Diligence, reproducibility, and agreement did not make the claim valid. The checkpointed run you just saw caught all three before a model was fit.

StatLab prompt-ablation study, Claude Opus 5, August 2026. Timing count from a search of each run's script and report.

I Do: what the agent did and what the analyst did

Agent contribution	Analyst responsibility
Inspected 1,254 rows and many variables quickly	Decided which discrepancies were scientifically consequential
Wrote and ran R code	Approved the formula, sample, and uncertainty method
Produced diagnostics and alternatives	Determined what each check could and could not establish
Saved reproducible outputs	Reframed the target when timing did not support causality
Drafted a report	Restricted the claim to an adjusted association

The junior ballgame is the entire analysis cycle with a smaller, inspectable decision space.

We Do: what does "salary difference" mean?

The data contain 397 faculty members at one US college in 2008–09.

Variable	Meaning
`salary`	Nine-month salary in dollars
`sex`	Female or Male
`rank`	Assistant, Associate, or Full Professor
`discipline`	Theoretical or applied field
`yrs.service`	Years at the institution
`yrs.since.phd`	Years since the PhD

Question for the room: A prompt says "Describe the adjusted Male–Female salary difference." Which decisions are still unresolved?

We Do: decisions the prompt leaves open

1. Adjust for discipline?
2. Adjust for experience—years of service, years since PhD, or both?
3. Adjust for academic rank?
4. Which interval, given 39 women and 358 men?
5. Describe the observed faculty or generalize beyond this college?

The data support descriptions of the overall disparity or of comparisons among faculty with similar observed roles. They do not identify a causal effect of sex.

There is no universally correct adjustment set. Each choice changes the question the coefficient answers.

We Do: adjustment changes the target



If rank partly reflects an institutional process of interest, adjusting for it removes that pathway from the coefficient. If the question is pay differences within rank, omitting it compares different roles.

The model cannot decide which institutional question matters.

We Do: what the brief fixes for every group

Each group starts from `exercises/we-do-salaries-analysis-brief.md`. These parts are given, so results can be compared at the end:

- **Data:** `data/salaries/salaries.csv`, read-only; all 397 rows, no missing values.
- **Contrast:** Male minus Female, Female as reference; salary in dollars, untransformed.
- **Estimator:** linear model with no interactions; report the `sexMale` coefficient.
- **Claim boundary:** this college in 2008–09; descriptive, not causal.
- **Workflow:** the agent audits and stops; it fits only after the group approves the plan.
- **Output:** `results.csv` with the same columns for every group, and a `submission.txt` line to paste into the class form.

We Do: what your group decides

Decision	Options
Comparison	Overall disparity, similar job context, or same rank and job context
Adjustment set	Discipline? Years of service, years since PhD, both, or neither? Rank?
Sensitivity model	None, or one alternative formula and the different question it answers
Uncertainty	Open: name a method, ask Claude for options and pick one, or write "Claude decides"

Write the target in one sentence in `brief.md` before the agent fits anything. Whoever chooses the uncertainty method, `report.md` must state what was done and who chose it.

We Do: run it with the agent

In groups, working in `scripts/we-do/<group-name>/`:

1. Pick a group name. Copy the brief and complete the decision sections.
2. Give Claude the brief. It audits and stops; read `audit.md`.
3. Approve or revise `plan.md`. Only then may it fit.
4. Run the script in a clean session. Read `report.md`: what was actually done?
5. On the class form, enter your group name and paste the contents of `submission.txt`.
6. Do Part A of the validation worksheet: check your own code and result. Part B waits for the comparison.



Class form

We Do: inspect your group's R

Read `analysis.R` without asking Claude first:

- What is the contrast and its direction?
- Which variables define the comparison? Do they match your brief?
- Which formula is primary, and which is sensitivity?
- Which rows enter each model?
- Which line computes the interval? Does it match what `report.md` says was done?

Syntax key: in `factor(x, levels = c("Female", "Male"))` the first level is the reference; `nobs(fit)` reports how many rows entered the model.

We Do: verify before interpreting

Check the shortest chain that could falsify the handoff:

1. **Data:** 397 rows, no missing values, 39 women and 358 men.
2. **Formula:** the formulas in the code match the brief.
3. **Reference:** `sexMale` means Male minus Female.
4. **Execution:** run the saved script in a clean R session.
5. **Independent check:** raw means reproduce the sign and $14,088 difference.
6. **Claim:** describes one college and year; does not assert causality.

```
with(d, mean(salary[sex == "Male"]) - mean(salary[sex == "Female"]))
# 14088.01
```

We Do: compare across groups

```
Follow instructor_files/compare-we-do-results.md
```

- Which groups answered the same question? Do their estimates agree?
- Where estimates differ, is the cause the target (formula) or the implementation (rows, reference level, interval)?
- Who chose each uncertainty method? Did groups that left it to Claude get the same one, and can they explain it?
- Did the method change which groups would call the difference clearly positive?
- Does each group's claim language match its formula?

We Do: the decision space

Facilitator-validated calculations, Male minus Female, all 397 rows:

`salary ~ sex +`	Estimate	Conventional 95%	HC2 95%
(nothing)	$14,088	4,131 to 24,045	5,327 to 22,849
discipline	$14,029	4,183 to 23,875	5,756 to 22,303
yrs.service	$9,072	−486 to 18,630	1,122 to 17,021
discipline + yrs.service	$8,423	−905 to 17,751	1,235 to 15,612
discipline + yrs.since.phd	$6,956	−1,909 to 15,821	779 to 13,134
rank	$4,943	−2,972 to 12,858	−737 to 10,623
rank + discipline + yrs.service	$4,771	−2,853 to 12,396	−88 to 9,631

These are not seven attempts to discover the "true" coefficient. They answer different descriptive questions.

Full grid: `scripts/we-do/salaries-reference-results.csv`. Intervals assume a repeated-process or superpopulation interpretation this single-college file does not establish.

We Do: a defensible handoff

For a group that chose discipline and years of service, and chose HC2 intervals itself:

> *In these 397 faculty records, men's mean nine-month salary was $14,088 higher than women's. Among faculty with the same discipline and years of service, the estimated difference was about $8,423 (HC2 95% interval $1,235 to $15,612). A model that also conditioned on rank estimated about $4,771, but that coefficient answers a different within-rank question. These observational data from one college and year do not identify a causal effect of sex.*

The handoff reports the planned primary result and labels the alternative rather than choosing whichever result is most convenient.

They Do: analyze ozone with a coding agent

On your own after the workshop: use this case to test how well you can run each step without scaffolding.

The data contain 153 consecutive days in New York, May–September 1973.

Target: the adjusted difference in mean afternoon ozone between 85°F and 75°F, standardized over observed wind, solar radiation, and month in the approved sample.

Boundary: a descriptive adjusted association, not a causal effect.

Known facts: ozone is missing on 37 days, solar radiation on 7, and nearby days may be dependent.

Your job is to direct a bounded analysis and decide what the result can support.

They Do: write before you run

Write a six-part brief in `scripts/they-do/<your-id>/brief.md`:

1. Goal and target contrast.
2. Data, unit, time order, and read-only boundary.
3. Primary estimator and exact analysis sample.
4. Required output files and one independent check.
5. Decisions the agent may not change without approval.
6. Evidence that triggers revise, stop, or escalation.

Ask the agent to audit first and stop. Approve or revise its sample and model plan before it fits anything.

They Do: questions to address or escalate

- **Coding:** Should month be categorical or numeric?
- **Support and sample:** Is the 85-versus-75 contrast supported within months, and what population does complete-case analysis describe?
- **Model:** Does the temperature relationship need a nonlinear term?
- **Uncertainty:** How should ordered daily observations affect the interval?
- **Verification:** Which check could reveal an implementation error?

Use Claude to inspect, calculate, and propose. Resolve what you can evaluate; record an escalation for consequential choices that exceed your expertise.

They Do: required handoff

Save four core artifacts:

- `brief.md`: your six-part prompt;
- `audit.md`: the agent's evidence about the data;
- `analysis.R` and `results.csv`: only if you approve an executable plan;
- `handoff.md`: the approved or rejected plan, one check and its limit, any unresolved decisions, and a bounded claim or escalation.

Also preserve the clean-session run log when code is executed.

Before finishing, you should be able to explain the formula, sample, contrast direction, and one verification result without asking Claude.

An incomplete analysis with a well-supported stop or escalation is a valid outcome.

A five-question review for any agent analysis

1. **Target:** What quantity and population does this result describe?
2. **Source:** Which consequential choices were specified, and which did the agent supply?
3. **Execution:** Does the code implement the approved sample, estimator, contrast, and uncertainty procedure?
4. **Evidence:** What check could expose an error, and what can it not establish?
5. **Boundary:** Can I evaluate this choice well enough to approve it?

If the fifth answer is no, narrow the task, learn the method, or seek expertise.

What to delegate—and what to retain

Usually productive to delegate	Retain or explicitly approve
Data inventory and mechanical summaries	Scientific question and target quantity
Draft R code for a specified estimator	Population, sample, and adjustment logic
Repetitive transformations and tables	Assumptions and uncertainty procedure
Clean-session execution and logging	Interpretation and strength of claim
Proposed diagnostics and alternatives	Whether a diagnostic is relevant and sufficient

Delegation is safest where the output is cheap to inspect and the consequence of error is understood.

TL;DR

Let the agent accelerate the parts of statistical analysis you can specify and inspect.

Do not delegate the meaning of the analysis beyond your ability to evaluate it.

A saved script answers "Can this computation be repeated?" Statistical judgment answers "Was this the right computation, and what does it warrant?"

Questions and application

Bring one analysis you might ask Claude to help with.

- What is the target quantity?
- Which part can be specified well enough to delegate?
- Which choice would exceed your present expertise?
- What artifact would make the work inspectable?
- What evidence would make you stop or reframe?

Appendix: research used in this workshop

- Alizadeh, M., Mosleh, M., Gilardi, F., Kasirzadeh, A., & Tucker, J. A. (2026). *AI Coding Agents Can Reproduce Social Science Findings.*
- Miao, J., Pritchard, J. K., & Zou, J. (2026). *The Agentic Garden of Forking Paths.*
- Rahman, M., et al. (2026). *DSAgentBench: Can Agents Automate End-to-End Data-Science Workflows in Real Computer Environments?*
- Sebo, P., & Wang, T. (2026). Evaluating the performance of ChatGPT-5, Claude, and DeepSeek for statistical analysis in medicine. *Internal and Emergency Medicine, 21*, 1227–1237.
- Song, X., et al. (2025). *Performance Evaluation of Large Language Models in Statistical Programming.*
- Zhu, Y., Ding, Y., Lai, P., Wang, L., Jing, B., & Chen, G. (2026). *StatABench: Dataset and Framework for Evaluating Statistical Analysis Capabilities of LLMs.*

Local evidence: StatLab prompt-ablation study, Claude Opus 5, August 2026 (21 runs). All claims are bounded to the model versions, prompts, harnesses, and tasks evaluated.